

Linkage Governance Framework

Current State Assessment Report

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¹ OS-PCORTF Project Information: <https://aspe.hhs.gov/creating-new-data-research-environment>.

Record of Changes

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2	May 2025	E. Kraus, G. Jenkins	Revisions based on feedback from sponsor team and CMS
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1 Introduction

1.1 Background

Linking individual health data across complementary data sources and connecting multiple data types increases the patterns and interrelationships that a researcher may explore to accelerate discoveries regarding prevention, diagnostics, and treatments.

The U.S. government (USG) aims to enhance the capability and capacity for data linkage in research by addressing a key challenge: governance. The current biomedical research infrastructure, including studies of cancer and patient-centered outcomes research (PCOR), lacks a common governance framework to support linking datasets for research. Researchers that seek to link datasets find themselves in an expensive and time-consuming process of attempting to align the governance of disparate datasets and harmonize their policy requirements.

A flexible and scalable common governance framework could be designed to enhance researchers' access to comprehensive and longitudinal individual-level data by streamlining governance processes and reconciling governance conflicts. **Flexible** governance is built to facilitate the harmonization of many data types and data collected for research and other purposes. **Scalable** governance is built to create multilateral linkages for research, public health, and population health applications drawing data from federal, academic, and other institutions. Multilateral linkages, where more than two datasets are linked, are an especially important capability that pushes the frontier of what is possible in terms of linking individual data for research. Unlocking linkage of three, four, or more datasets would help researchers construct the most complete picture of an individual's health and conduct research less impacted by statistical bias caused by missing or incomplete information.

The USG initiated a project with the Health Federally Funded Research and Development Center (FFRDC), operated by MITRE, to develop and pilot a common governance framework to advance data linkage for research. To start this design process, MITRE conducted a current state assessment of three governance models that successfully support linkage from 12 existing data systems, nominated by a U.S. Department of Health and Human Services (HHS) working group. Each system and governance model was analyzed to consider what lessons learned might contribute to a common governance model. This document shares the findings of this current state assessment.

1.2 Purpose

The purpose of the current state assessment was to:

1. Describe three governance models for linking data and providing linked data to researchers.
2. Evaluate each model's efficiency, flexibility, and scalability.
3. Identify model attributes that would limit its flexibility and scalability.

Findings from the current state assessment informed an options memo developed by MITRE and submitted to the USG titled *Options to Enable Flexible, Scalable, Linked Data for Research*.

1.3 Audience

The primary audience for this current state assessment is the National Institutes of Health (NIH) National Cancer Institute (NCI, the project sponsor), which seeks to use these findings to improve governance for linkage of multiple datasets for research. Secondary audiences include other Institutes, Centers, and Offices at NIH as well as other HHS agencies that are engaged in the project's Executive Steering Committee and provide expert advice to this project and other USG partners engaged in relevant linkage or research governance activities.

2 Methods

The current state assessment included a comprehensive review of three data systems that facilitate linking individual-level data for biomedical research.

The Health FFRDC convened a project team and conducted the following activities as part of the methodology of the current state assessment:

- Created a data collection tool,
- Identified and accessed technical and governance materials relevant to each data system and requested additional materials not publicly available,
- Reviewed all materials, extracted governance content, and entered it into the data collection tool,
- Interviewed representatives from each data system,
- Mapped the governance process to request data from each system and validated a process map describing each data system, and
- Documented the level of effort for each data system's governance process.

Though the current state assessment broadly examined each data system, it primarily focused on governance policy and process documentation, governance agreements, and technical documents relevant to research and linkage.

To supplement learnings from the examination of three data systems, experts in data sharing, biomedical research, and data linkage were also interviewed.

2.1 Data System Selection

The USG identified 12 candidate data systems and approaches for inclusion in the current state assessment:

- Chronic Conditions Warehouse (CCW) Virtual Research Data Center (VRDC, under the Centers for Medicare & Medicaid Services (CMS))
- National Clinical Cohort Collaborative² (N3C, under the NIH National Center for Advancing Clinical and Translational Sciences or NCATS)

² Formerly known as the National COVID Cohort Collaborative.

- The Sentinel Program (under the Food and Drug Administration)
- National Patient-Centered Clinical Research Network (PCORnet)
- The Surveillance, Epidemiology, and End Results (SEER) Registries (under NCI)
- *All of Us* Research Program (under NIH)
- Veterans Health Administration Corporate Data Warehouse
- Centers for Disease Control and Prevention National Program of Cancer Registries
- National Heart, Lung, and Blood Institute BioData Catalyst Repository (under NIH)
- HHS Protect
- Trusted Exchange Framework and Common Agreement (TEFCA)
- National Strategic Computing Reserve

To select three candidates for inclusion in this assessment, USG evaluated whether each data system generates data for research, whether each data system has a governance model managing a large volume and variety of data, and whether the data system supports linkage at the individual level. Candidates that did not meet all three criteria were excluded from the current state assessment and among the candidates that did meet all the criteria, those with the most governance maturity (e.g., longevity, established policies and procedures, and available documentation) were prioritized.

After evaluating candidates, USG selected three data systems for the current state assessment: [N3C](#), [PCORnet](#), and [VRDC](#). N3C was selected as a linked data system with a high-functioning governance model based on a data enclave approach. PCORnet varies widely in its approach to data linkage and was selected as a federated data exchange approach with a mature governance model. VRDC was selected as a high functioning governance model and system to provide CMS data for research and how it exemplifies a federal agency's approach to linking data and making it available for research.

2.2 Data Collection

For each data system, the assessment sought to understand:

- The data system's technical approach to collecting, storing, linking, and sharing data for research
- How the data system is currently used by researchers (e.g., how many studies, what types of studies, what types of researchers)
- What types of and how much data the system holds or manages
- The mechanics of how data are linked internally (within the system) and externally (internal data linked to external datasets)
- The governance policies, processes, and agreements that support:
 - System operations
 - Data contribution
 - Data linkage
 - Researcher data request and use

- The data system's technical and governance approach to ensuring security and privacy
- The legal and security requirements to which the data system must adhere and how the system meets said requirements

The current state assessment's data collection activities were designed to summarize and synthesize key information rather than reproducing each data system's extensive documentation.

Data System Material Collection and Review

For each data system, the project team collected materials by:

- Reviewing each data system's website(s) and downloading all hosted documents and linked content (with a focus on more recent content)
- Conducting a scan of academic literature to identify peer reviewed work published by the data system or using data from the system (with a focus on more recent publications)
- Requesting additional materials not publicly available directly from the data system

The project team collated all materials in a worksheet and mapped each item to a set of system and governance topics to guide team members who were assigned to specific topics on which materials to review.

The project team tracked which items were reviewed to ensure that every item was read by at least two team members.

Data Collection Tool

The project team created a data collection tool template and drafted over 90 questions (see [Appendix A](#)) to address key assessment topics. Information gathered from the data system materials was documented in the data collection tool and organized by the following topics:

- Technical overview
- Governance overview
- Data use
- Linkage and linked data
- Governance for requesting data for linkage
- Human subjects research protection and ethics
- Privacy board, privacy and security
- Data use agreement
- Laws
- Other governance (unique aspects of the governance model not addressed in any other section)
- Governance experience (qualitative questions about how well the governance model was meeting the system's and user's needs)

The project team replicated the template for each data system and customized the data collection tool by adding questions to capture unique aspects of the governance model,

nuanced governance information, and other notable findings. The project team worked on one data collection tool for one system at a time. One or two team members were assigned to review materials for each topic above and answer questions in the data collection tool based on information from each data system. When reviewing materials and entering content into the data collection tool, sources for each entry were cited. The project team also flagged questions and content that required clarification. When appropriate, direct excerpts (e.g., exact policy language or quote from article) were inserted into the data collection tool for reference. Completing the data collection tool helped identify gaps in critical information that could only be filled through interviews (see [Section 2.4](#), below).

The project team generated a comprehensive list of all reviewed material for each system for reference (see [Appendices B, C, and D](#)).

2.3 Governance Process Map

Using the data system materials, the project team mapped the process a researcher follows to request data from each of the three data systems, including activities by the researcher and various other entities like the Institutional Review Board (IRB), Privacy Board, reviewers, and contract office. The purpose of the process maps was to visualize in a comparable way the process to initiate linkage for research across data systems. The process maps allowed for comparison of the sequence and similarity of process steps, decision makers, and decision points. The project team validated the process maps with representatives from each data system during interviews, updating the process maps and presenting them again at a subsequent interview or confirming their accuracy over email.

2.4 Data System Interviews

The project team conducted interviews with leaders from each of the three data systems to:

- Confirm the interpretation of reviewed materials,
- Clarify content from reviewed materials that was unclear or contradictory,
- Answer questions in the data collection tool where no explanation could be gleaned from the available materials,
- Validate the governance process map, and
- Discuss how well the governance model operates and any plans to alter or change it.

Interviews were scheduled for one hour, or more if needed, and were held after the project team had completed the review of data system materials and had populated the data collection tool.

2.5 Targeted Feedback

The project team also conducted six targeted feedback sessions with subject matter experts from several HHS agencies to discuss themes arising from the current state assessment and align those themes with the larger data sharing, linkage, and research ecosystem.

2.6 Analysis and Synthesis

Once the data collection and interviews were completed, the project team drafted a summary of findings for each data system and synthesized findings to discuss limitations in the governance model's ability to scale, flexible features, scalable features, and efficiencies based on the following definitions:

- **Efficiency** is the speed from researcher request to data access, the fewest possible process steps and required approvals, and higher throughput.
- **Flexibility** is built to facilitate the harmonization of many data types and data collected for purposes other than research.
- **Scalability** is built to create multilateral linkages for research, public health, and population health applications drawing data from federal, academic, and other institutions.

Findings were then compared across models to identify common themes for this report.

The USG reviewed draft summaries, providing feedback that the project team incorporated into updated summaries. Each federal program interviewee subsequently reviewed and approved their respective summaries to ensure agreement.

3 Findings

3.1 National Clinical Cohort Collaborative – COVID-19 Enclave

The project team conducted three interviews with N3C partners and analyzed 35 resources (see [Appendix B](#)). Though N3C has recently expanded beyond COVID-19, the project team limited its review of materials and analysis to the COVID-related infrastructure, governance, and functions.

Summary

NCATS created the National COVID Cohort Collaborative in 2020 in response to the COVID-19 pandemic, establishing the nation's largest open-science resource of real-world data for COVID-19. Renamed the National Clinical Cohort Collaborative in 2024, N3C is a technology partnership that assembles health data about COVID-19 and other clinical conditions for research and public health purposes.

N3C is fundamentally premised on a centralized architecture that accepts, links, harmonizes, and stores data provided by data contributors in a secure enclave for research and public health. N3C's governance is structured to reflect specific values and a vision for research, including accessibility of data, promoting team science, and building trust among data contributors through transparency and collaboration. The NCATS N3C leadership team provides governance for the N3C system in coordination with the NCATS Division of Clinical Innovation. The governance described below is limited to N3C's COVID-19 data policies and processes. N3C's tenant and freemove models are described separately at the end.

N3C's primary goal was to harness clinical data to better understand, treat, and mitigate the effects of COVID-19. Created to enable large-scale, real-time research on COVID-19, users can access data through N3C to study risk factors, treatment efficacy, disease progression, and long-term effects. NCATS designed N3C based on the following decisions: 1) the architecture will reflect a centralized design, 2) researchers will only access data through a secure enclave, 3) the enclave will be government-owned and -operated, and 4) technical partners that help operate N3C will be supported with contracts and not grants. To develop N3C, NCATS collaborated with the Clinical and Translational Science Awards (CTSA) Program awardees, and other academic institutions, healthcare providers, and federal agencies to co-design the technology and governance that make data contribution, data harmonization, and data access possible.

N3C's government-owned technical architecture is free to use and is operated by NCATS with support from contracted commercial partners including Palantir (Foundry analytics platform), Regenstreif Institute (linkage honest broker) and Datavant (linkage software). NCATS estimates the current cost for operating N3C at roughly \$25 million per year.³ Over 80 data contributors, including most CTSA awardees,⁴ provide data every week that includes COVID patients, control groups, and those having received COVID-19 vaccines. Thus, data within N3C is not static because the data on patients whose records are deposited in N3C grows as they receive more healthcare.

The N3C governance model and technology has been highly successful, informing research on COVID-19 epidemiology, clinical outcomes including long COVID, treatment efficacy, and population and genetic risk factors. To date it has provided data to over 4,000 users, leading to over 300 scientific publications. The N3C data journey begins with data contributors extracting individual-level data on COVID-19 patients and non-COVID controls from electronic health record (EHR) systems. Data contributors may submit data to N3C in the PCORnet, Observational Medical Outcomes Partnership (OMOP), TriNetX, or Accrual to Clinical Trials Network data model format (most data contributors already have at least one of these data models built). Data contributors extract and transmit data via weekly secure file transport protocol to N3C. Depending on the contents of the data, the transmitted datasets may include location information and dates, meeting the definition of a limited dataset. N3C relies on data contributors to identify unique individuals consistently within their datasets using a common identifier. N3C uses the OMOP data model internally, mapping data received from all other formats to the OMOP model and generating new data releases for research users each week. Over 30 billion rows of health data representing about 23 million individuals are held by N3C as of fall 2024.

N3C's governance reflects a decentralized model. Four governance workstreams are responsible for different governance domains, and each has a workgroup that meets regularly and is open to participation. N3C data contribution is governed by a Data Transfer Agreement (DTA) and a Linkage Honest Broker Agreement (LHBA) while data use is governed by an institutional Data Use Agreement (DUA), a Data User Code of Conduct, a set of Community

³ A 2024 U.S. Government Accountability Office audit estimates N3C's FY2023 costs at \$50 million. <https://www.gao.gov/assets/gao-24-106638.pdf>

⁴ According to NCATS, each participating CTSA received \$100,000 to support N3C data contribution.

Guiding Principles, and a Data Use Request (DUR) form and review process. These documents were developed by the governance workgroups through an iterative process that engaged multiple partners.

The N3C DTA governs the contribution of data to N3C's data system. The DUA governs data use at the institutional level (not the project level). Notably, the terms in both the DTA and DUA are non-negotiable: fixed terms ensure that N3C does not have to tag data with computable metadata and rules to enforce corresponding agreement terms. Non-negotiable DTAs and DUAs also support the transparency valued by N3C, ensuring that all institutions know they are being treated the same way.

Aligned to its values, N3C data may be used by researchers ranging from federally funded researchers to citizen scientists (for whom no institutional affiliation is required). Citizen scientists may elect to use synthetic data or gain access to more sensitive data by partnering with a research institution or joining an existing research team. A researcher's institution must execute an institutional DUA before the data request process may begin. The DUA parameters include the N3C Data User Code of Conduct, which restricts data use to COVID-related projects, establishes an expectation of confidentiality, requires compliance with applicable laws and reporting of data access incidents, mandates appropriate N3C acknowledgment and attribution, and prohibits identification or inferences of patients, institutions, or communities (including tribal affiliation). N3C has developed and published extensive governance documentation to the web to support users and drive research. Publications using N3C data are bound by the N3C Attribution Policy that requires proper attribution and encourages open access publication.

Researchers from organizations with an institutional DUA may request a limited or deidentified dataset through a centralized DUR process. DURs contain only descriptions of the desired dataset, research aims, and the benefits of the research as they relate to COVID. To submit a DUR, users must have an Open Researcher and Contributor ID or ORCID (a unique persistent identifier for individuals to use as they engage in research⁵) and register with the N3C COVID Data Enclave (the Enclave). Following submission, DURs are reviewed by two members of the N3C Data Access Committee⁶ (DAC) in a matter of days. The DAC typically grants DUR approvals in less than one week. Projects approved through N3C's DUR process are posted publicly⁷ to encourage collaboration and team science. NCATS staff then create a project-specific workspace within the Enclave that contains the dataset that the researcher requested and grants the researcher access. In terms of governance enforcement, NCATS has only found it necessary to take an enforcement action on one occasion to date. N3C's primary enforcement mechanism is to revoke a researcher's Enclave access with the potential to escalate by revoking access for all users from the researcher's institution.

The Enclave is the sole location where N3C data may be accessed and is a critical part of N3C's and NCATS' approach to ensuring data privacy and security. Individual-level data does not leave the Enclave. To ensure that no patient can be reidentified, researchers are permitted to download only aggregate results (with no counts under 20 persons) from the Enclave and

⁵ ORCID: <https://orcid.org>

⁶ The DAC is composed of representatives across NIH and representing various research domains.

⁷ N3C Approve Projects: <https://covid.cd2h.org/projects/>

only after a review by the Results Download Review Subcommittee (per the Results Download Policy). As a government-owned system, the N3C technical architecture is subject to the Federal Risk and Authorization Management Program (FedRAMP).⁸ NCATS has assessed N3C as a FedRAMP “moderate” risk system but plans to address risk at the “high” level in the future. NCATS has also executed a Privacy Impact Assessment (PIA) on N3C, although the PIA has not been made public.

A variety of federal laws and regulations protect N3C data, including the Privacy Act of 1974 and the Health Insurance Portability and Accountability Act (HIPAA) Privacy and Security Rules and most N3C data contributors are HIPAA covered entities. Therefore, N3C has created governance processes that comply with the Privacy Act and the HIPAA Privacy and Security Rules including maintaining limited and deidentified versions of datasets for use. N3C defines limited and deidentified datasets according to HIPAA, using date shifting to retain dates in a deidentified dataset.

N3C governance addresses human subjects protections review for data contribution and data use. To minimize redundant review processes across multiple institutions, N3C employs a centralized IRB model for data contribution.⁹ The N3C technical architecture and data contribution process have been reviewed by a lead IRB and determined to be of minimal risk to patients; the lead IRB granted a waiver for the requirement of informed consent based on that determination and the nature of the project. Participating institutions have the option to cede IRB authority to the lead IRB overseeing the N3C protocol. If data contributors choose to conduct their own IRB review, they can use a curated IRB protocol provided by N3C specifically designed for data submission that ensures alignment with N3C's governance and regulatory requirements. For requests of limited datasets, N3C requires researchers to gain approvals from their local IRB, aligning with the requirements of the HHS regulations for the protection of human subjects in research at 45 CFR 46 (hereafter 45 CFR 46, which has five parts including Subpart A, known as the Common Rule).¹⁰ Users accessing a deidentified N3C dataset are not required to complete IRB review. Notably, N3C does not receive data collected as part of research and subject to a research consent, eliminating the need to align N3C operations with terms across multiple different consent documents.

N3C uses privacy-preserving record linkage (PPRL) via tokenization for three linkage activities: deduplication, linkage to external datasets, and cohort discovery. PPRL participation is voluntary and for all those who participate, N3C links and removes duplicate patients within and across data contributors. N3C uses a linkage software and Linkage Honest Broker (LHB) to operate PPRL; N3C currently contracts with Datavant and the Regenstrief Institute (Regenstrief) for their linkage software and LHB services, respectively. PPRL participation requires data contributors to sign the LHBA with both NCATS and the LHB; that agreement formalizes the terms of PPRL. Data contributors are also required to make elections regarding their participation in linkage to external datasets and cohort discovery, which are optional linkage

⁸ From the 2024 GAO Audit, “the National COVID Cohort Collaborative is a subsystem under the National Center for Advancing Translational Sciences General Support System, which has a privacy plan that includes privacy controls and requirements.” <https://www.gao.gov/assets/gao-24-106638.pdf>

⁹ As of fall 2024, the Johns Hopkins IRB is the lead IRB.

¹⁰ HHS regulations for the protection of human subjects in research at 45 CFR 46: <https://www.hhs.gov/ohrp/regulations-and-policy/regulations/45-cfr-46/index.html>

activities. Every data contributor who participates in PPRL installs Datavant software to generate tokens. Datavant carries an active expert determination that Datavant-generated tokens are deidentified under the HIPAA Privacy Rule standards. N3C states that the PPRL solution is vendor neutral such that an organization other than Regenstrief and software other than Datavant could be used.

N3C allows linkage to external datasets, meaning linking patients in a dataset that is not being contributed to N3C to individuals within N3C contributed datasets. Researchers can also request that publicly available datasets be ingested into the Enclave in order to link those publicly available datasets to the N3C data and make linked data available for use by researchers in the Enclave.¹¹ For external datasets, N3C uses the same PPRL method and partners described above. However, fewer than ten external datasets have been linked to date and this process is driven by NCATS, not researcher initiated. To determine which external datasets may be linked, N3C classifies external datasets based on risk of reidentification and links only datasets that do not increase risk (Table 1).

Table 1. National Clinical Cohort Collaborative External Dataset Linkage Classification Summary

Class Number	Linkage Classification
Class 0	Linkages using cryptographic hash codes (tokens) managed by a third-party linkage honest broker to connect multiple enclaves
Class 1	Linkages leading to immediate reidentification of patients and not permitted with N3C
Class 2	Linkages using cryptographic hash codes (tokens) within a single enclave leading to higher confidence reidentification of patients
Class 3	Linkages leading to data sufficiently aggregated as to reasonably mitigate the risk of reidentification
Class 4	Linkages or use of data not involving individual persons

For class 0, 2, 3, and 4 datasets, NCATS provides a Datavant license to the dataset holder to generate and send tokens to the LHB to create linkage maps. NCATS then generates a static linked dataset that contains N3C data with the external dataset for use. Because NCATS must purchase the Datavant license for each dataset holder, NCATS carefully selects for linkage external datasets that have enduring value for many users. NCATS hopes to expand this functionality to allow researchers to request linkage of external datasets on a project basis.

¹¹ Publicly Available External Datasets: <https://covid.cd2h.org/external-datasets/>

Alzheimer's, Renal, and Chronic Obstructive Pulmonary Disease Tenant Pilots

Building on the success of N3C, NCATS has developed and begun testing a tenant model to extend N3C beyond use cases and datasets relevant to COVID.^{12,13} A phase-one pilot study has tested the tenant model's capacity to expand to three non-COVID conditions: Alzheimer's disease and related dementias, renal disease, and chronic obstructive pulmonary disease. The tenant model requires data contributors to provide a larger, disease-agnostic sample of EHR data (more data fields and a larger clinical population than N3C collects for its COVID Enclave), from which NCATS generates "a tenant" in an enclave.¹⁴ A tenant is a temporary secure enclave where NCATS has pulled topic-specific data on a specified phenotype. Conceptually, NCATS provides the access and computing structure (metaphorically "the condominium") and the researcher and research topic data are moved into the condo (as a "tenant")—providing access only to authorized users. Once the research concludes, the tenant enclave and its data sunset. The agreements and governance framework for the tenant model have not been tested at scale but have potential as both pilots have thus far successfully demonstrated the proposed data flow.

Enhanced Linkages and N3C Freeport

In response to barriers in enclave-to-enclave data sharing, NCATS initiated two phase-two pilots exploring both the use of enhanced data linkages and a freeport concept, funded by the Office of the Secretary PCOR Trust Fund (OS-PCORTF). The first, *Capturing the Full Trajectory of Patient-Centered Cancer Care Via Enhanced Data Linkages*,¹⁵ is in partnership with the NCI. Diseases like cancer remain a major health challenge in the United States, but fragmented data across registries and electronic health records hinder a comprehensive understanding of patient care and treatment outcomes. To address these gaps, a cancer tenant aims to enhance data capacity for patient-centered outcomes research by linking NCI's SEER cancer registries with EHR data from NCATS' N3C. By evaluating these linked datasets through treatment cohort analyses for both common and rare cancers, the project will enable researchers to better track cancer patients' care trajectories and assess treatment effectiveness.

The second, *N3C Renal Freeport: A Collaborative Data Analytics and Data Sharing Model for Patient-Centered Outcomes Research* project,¹⁶ in partnership with National Institute of Diabetes and Digestive and Kidney Diseases, focuses on demonstrating a freeport concept for

¹² Suchetha Sharma, Jiebei Liu, Amy Caroline Abramowitz, Carol Reynolds Geary, Karen C Johnston, Carol Manning, John Darrell Van Horn, Andrea Zhou, Alfred J Anzalone, Johanna Loomba, Emily Pfaff, Don Brown, Leveraging multi-site electronic health data for characterization of subtypes: a pilot study of dementia in the N3C Clinical Tenant, *JAMIA Open*, Volume 7, Issue 3, October 2024, ooae076, <https://doi.org/10.1093/jamiaopen/ooae076>

¹³ N3C Tenant Report: <https://zenodo.org/records/10368895>

¹⁴ N3C Tenant GitHub: <https://github.com/National-Clinical-Cohort-Collaborative/tenant-pilot/wiki/>

¹⁵ Capturing the Full Trajectory of Patient-Centered Cancer Care Via Enhanced Data Linkages Project Information: <https://aspe.hhs.gov/capturing-full-trajectory-patient-centered-cancer-care-enhanced-data-linkages>

¹⁶ N3C Renal Freeport funding information: <https://aspe.hhs.gov/n3c-renal-freeport-collaborative-data-analytics-data-sharing>

renal health. N3C and many others restrict data access within an enclave, which means that research seeking multilateral linkages of data that are confined to secure enclaves is not possible. Governance is needed to support research that requires connections across multiple enclaves. In such a scenario, N3C imagines that researchers may use a mutually agreed-on temporary workspace provided by a secure, neutral cloud-based enclave (a freeport) to connect multiple existing enclaves. Freeports provide a model for ensuring reliable and safe data sharing between agencies and enclaves. Essential to the freeport is a governance mechanism to formalize mutually agreed-on security and privacy parameters. N3C has piloted the use of an Interconnect Agreement to specify privacy and security parameters across tenants. The freeport supports linkage by including the LHB on the Interconnect Agreement and allowing the LHB to deposit linkage maps into the freeport. Conceptually, a freeport is accessible by all authorized users and owned by no one.

Limitations in N3C's Governance and Linkage Architecture

- N3C governance for COVID operates for the contribution and use of COVID data only. While the tenant pilots experiment with how governance for COVID can flex across other health domains, the tenant governance model is still under development.
- The amount of decision making that data contributors have delegated to NCATS is unusually large and may reflect the narrowness of COVID data and/or a collaborative inertia generated by the COVID-19 pandemic response.
- CSTA Program awardees are the primary contributors of clinical data and are awarded funds to support their N3C data contribution. While N3C participation is voluntary, CSTA awardees are incentivized to provide data. The degree to which holders of data would agree to the N3C architecture and governance without such financial support is unclear.
- N3C states that its linkage solution is vendor neutral; while the governance approach for linkage may be transferable away from Datavant to another vendor, the level of effort required to change linkage software vendors would be significant.
- N3C's approach to PPRL carries substantial costs where software fees increase for every organization providing data for linkage. The N3C linkage approach has not created a scalable solution for researcher-initiated linkage of external datasets.

Efficiency in N3C's Governance and Linkage Architecture

- N3C governance is well documented and transparent. NCATS provides clear and widely available instructions on how N3C should be used, steps to gain access, and expectations of users. The result has been thousands of users and uses and an impressive trajectory of growth and expansion.
- Requiring a single institutional DUA for multiple institutional users reduces the number of DUAs and the burden of negotiating the DUA, reducing start-up time for subsequent requests from that institution.
- N3C governance provides quick access to data because the DUR process is simple and fast. There is one approval to gain access to N3C data and the N3C team is dedicated and resourced to provide rapid reviews.

- Data contributors do not opt in or out of each project; data contributors allow the DUR process to determine which projects permit access to N3C data and that negates the need for an additional participation process that slows data access.
- N3C governance has streamlined review of human subjects protections on both data contribution and data use.
 - Data contributors either cede to the lead IRB or reuse a curated IRB protocol for submission.
 - Use of deidentified datasets does not require an IRB review or approval.
- By using date shifting to retain dates in a deidentified dataset, N3C has made deidentified datasets more usable and desirable.
- By using Datavant to generate tokens, N3C eliminates the requirement to conduct expert determination reviews on the deidentification of tokens because the Datavant PPRL platform ensures compliance with deidentification standards without requiring additional "expert determination" reviews for the deidentification of tokens used in the linkage process.

Flexibility in N3C's Governance and Linkage Architecture

- N3C ingests many different types of data (EHRs, claims, vaccine records, COVID test results, and social determinants of health) and accepts submissions across multiple different data models, so many different types of data contributors can submit data to N3C and studies on many different COVID subtopics are possible.
- N3C's institutional data use agreement does not include study-level detail so that one DUA can cover many investigators and different types of studies.
- N3C's architecture is not research only—it actively supports both research and public health initiatives related to COVID-19.
- N3C places near zero restrictions on allowed types of research studies so long as the researcher can demonstrate a plausible relation to COVID and potential future benefit to COVID patients.
- Linkage occurs through PPRL, which can be used across data types and requires only the availability of sufficient quality identifiers and ability to install software.
- Participation in PPRL is voluntary and data contributors can participate at multiple levels.
- N3C has navigated handling and provisioning access to both linked and unlinked data as not all N3C data contributors participate in linkage.

Scalability in N3C's Governance and Linkage Architecture

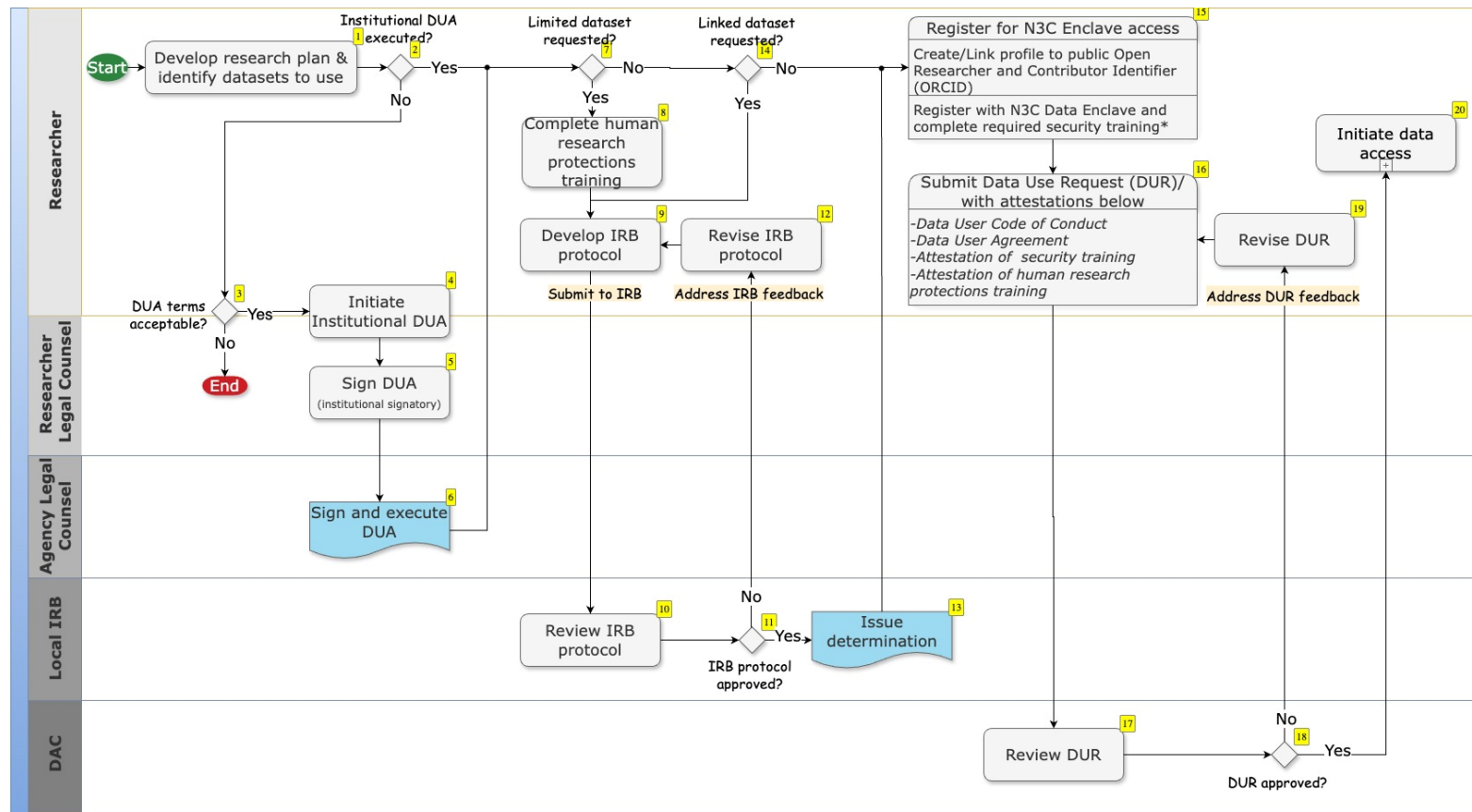
- N3C offers a single secure location where researchers access data so that the potential security and privacy points of failure are limited. This technical approach also instills trust with data contributors and minimizes time to retrieve data for a request.
- A simple and well-documented process and single approval permit access to a range of data types quickly.
- Participation in N3C's governance via workgroups is open and optional.

- N3C uses non-negotiable DUAs, as well as a simple proposal process, both of which streamline the approval process and improve scalability.
- Holding data centrally allows N3C to link all incoming datasets and refreshing those matches weekly provides a model for multilateral linkage.
- N3C's technical architecture and governance reflect its values of transparency, collaboration, trust, and team science, which have translated well to acceptance by a broad community.
- N3C's governance model relies on one simple and fast review and approval process to permission access to N3C data where two individuals on the DAC review a DUR and when approved, provides access to linked dataset.
- Accepting and harmonizing data from multiple common data models helps data contributors participate with limited development work.

Governance Process

Figure 1 displays the current N3C governance process that researchers use to request N3C data.

Figure 1. National Clinical Cohort Collaborative Governance Process to Request Access to the COVID-19 Enclave



* Required Security Training: Information Security, Counterintelligence, Privacy Awareness, Records Management Refresher, Emergency Preparedness Refresher

3.2 Virtual Research Data Center

The project team conducted two interviews with VRDC partners and reviewed 29 resources (see [Appendix C](#)).

Summary

CMS CCW VRDC provides approved researchers with direct access to Medicare and Medicaid data.¹⁷ The CMS Office of Enterprise Data and Analytics (OEDA) manages VRDC, a secure virtual enclave environment where researchers can access and analyze CMS data. A multidisciplinary team coordinates the governance process, creates datasets, and supports users. In order to receive access to VRDC, researchers must submit request paperwork to CMS, receive approval from the CMS Privacy Board, sign a DUA, and pay fees equal to the cost of making the data available.

Researchers associated with various types of organizations, including federal and state agencies, academic organizations, and for-profit organizations, may access CMS data in VRDC. Data accessed through VRDC can be used only for research purposes. The data that researchers may request in VRDC includes:

- Medicare Beneficiary Enrollment Data including demographic and chronic condition information on Medicare enrollees
- Medicare Fee-for-Service Claims (Parts A and B) Data on care provided to Medicare enrollees by hospitals, clinics, skilled nursing facilities, home health agencies, and other providers
- Medicare Part C Encounter Data from Medicare Advantage plans
- Medicare Part D Data from Prescription Drug Plans and Medicare Advantage Prescription Drug Plans
- Assessment Data from patient assessments conducted in care facilities
- Medicaid Eligibility, Enrollment, and Utilization Data from state Medicaid programs
- Data from Special Programs such as Center for Medicare & Medicaid Innovation Model Data Sharing

CMS maintains three categories of data files: Research Identifiable Files (RIFs), Limited Data Set (LDS) files, and Public Use Files (PUFs). The data available in VRDC are RIFs, which have the highest level of sensitivity since they contain protected health information and may contain limited direct identifiers. VRDC does not support requests for PUFs and LDS.

CMS also permits researchers to link external datasets to CMS data at the individual level using RIFs. As this report is focused on linkage, the discussion that follows is focused on how researchers apply for RIFs for access within VRDC.¹⁸

¹⁷ Established by the Medicare Modernization Act of 2003.

¹⁸ While CMS does allow researchers to request access to data on physical media extracted from VRDC, accessing data within VRDC is the preferred method because it limits privacy and security risk.

Governance

A variety of federal laws and regulations protect CMS data, including the Privacy Act of 1974 and the HIPAA Privacy and Security Rules. The Director of the CMS Office of Enterprise Data & Analytics and CMS Chief Data Officer (CDO) oversee governance and operations of the process to request RIFs. Governance documentation is developed and maintained by OEDA. All governance documentation is publicly available on the CMS and Research Data Assistance Center (ResDAC) website. ResDAC is CMS's technical assistance contractor that supports researchers in completing the necessary steps to receive access to the RIFs.

Request Process

Researchers begin by developing a RIF request package that includes:

- Application including study aims, data being requested, and proposed linkages to other datasets
- IRB documentation (that meets the requirements of the Common Rule and HIPAA)¹⁹
- Key Personnel Supplement
- RIF Specifications Worksheet
- RIF Data Use Agreement
- Collaborating Organization Supplement, if needed

ResDAC facilitates RIF request submission to OEDA for review and approval by the CMS Privacy Board. The CMS Privacy Board, comprising approximately 30 individuals across CMS, safeguards personally identifiable information and protected health information (PHI) by ensuring that disclosures of CMS data for research purposes comply with applicable laws and policies. Specifically, the Privacy Board focuses on evaluating whether the requested data is the minimum necessary for the research project and is appropriate for the stated research purpose and that the plan for dissemination of research findings is adequate. In general, the research request process relies on one review and decision-making body. Special requests (e.g., other federal agencies requesting non-standard files) may require additional review.

Once a request is approved by the CMS Privacy Board, CMS notifies the researcher, processes the RIF DUA, and requests payment.²⁰ The terms in the RIF DUA are non-negotiable and include specific requirements regarding data use as well as privacy and security standards.

Once payment is received, CMS onboards the researcher to VRDC, and provisions access to the files listed on the RIF DUA. Before researchers in VRDC can download analyses, CMS performs an output review to ensure that all information is deidentified according to the HIPAA Safe Harbor standard and meets the CMS cell suppression policy.

Security

¹⁹ Researchers must submit their to study an IRB before requesting data from CMS.

²⁰ The researcher must deliver payment before a dataset is prepared, or a researcher gains access to the data via VRDC. As CMS is a government agency, researchers tender payment to the U.S. Treasury.

VRDC's enclave uses a virtual cloud-based infrastructure that includes a virtual desktop interface managed by CMS with multi-factor authentication. When accessing data through the enclave, researchers must complete onboarding and training including identity verification and system orientation. Researchers are assigned individual user accounts and are prohibited from sharing access. The workspace provides a set of software and tools pre-approved for use (SAS, Databricks, STATA, etc.). It is designed to handle a wide range of analytical tasks while maintaining strict controls over the data's location and accessibility.

While certain researchers can elect to receive data on physical media, researchers are encouraged to use VRDC. Use of VRDC eliminates the risk of the loss of physical data and allows CMS to implement technical controls that address the HIPAA Privacy and Security Rules.

CMS continuously monitors VRDC user activity within the enclave to detect unusual activity or attempts to bypass security measures. The virtual desktop infrastructure is updated regularly to include the latest security patches and enhancements. CMS monitors advancements in data security technologies and continuously updates its protocols to address new threats and vulnerabilities, ensuring VRDC remains a trusted resource for health services research. Security decisions and operational changes affecting VRDC are primarily directed by the CDO.

Linking Data

Within VRDC, CMS assigns each individual a unique and persistent beneficiary identifier that allows an individual's data to be linked across CMS data sources and years. Researchers may request linkages to other non-CMS data sources by providing a description of the data and the linkage methodology in their RIF request package.

Requests to link individual-level data with CMS data will be approved only if CMS performs the match. Researchers must submit a finder file with identifiers from the other data source to CMS. Researchers can submit one primary identifier or at least three secondary identifiers to CMS to support data linkage. Examples of primary identifiers include Social Security Number (SSN), Health Insurance Claim Number and/or Medicare Beneficiary Identifier, and Medicaid Statistical Information System ID. Secondary identifiers include last name, gender, date of birth, ZIP code, and partial SSN.

CMS is partnering with NCATS to test linking data in N3C with CMS data using tokenization where the exposure and exchange of identifiers could be reduced. In this pilot approach, CMS would transfer tokens generated by Datavant software to the N3C LHB, Datavant would return matches and unique identifiers for matching individuals, and CMS would then prepare and transmit a file containing only matching individuals to N3C.

Cost and Funding

CMS is permitted under Section 1106(b) to recoup the cost of making the CMS data available. CMS does this through data fees. Fees vary based on the number of VRDC users and the tools they are using in VRDC (e.g., SAS-only or Databricks).

Limitations in VRDC's Governance and Linkage Architecture

- CMS's use of a single board review is appropriate for its data and demand signals but may be challenging to scale.
- CMS established only one method for external dataset linkage and allows only one CMS contractor to generate a linked dataset.
- CMS data fees may pose financial barriers for underfunded researchers or institutions.

Efficiency in VRDC's Governance and Linkage Architecture

- ResDAC's support ensures that most requests that undergo CMS Privacy Board review are complete and have a high likelihood of approval. ResDAC assists researchers in describing proposed research and linkage with sufficient clarity and detail for the research request to be approved.
- CMS allows researchers to amend or extend their research request so they do not have to repeat the governance process to modify their study.
- VRDC consolidates all necessary tools, software, and data in one secure location, simplifying workflows for data access.
- For federal agency researchers, CMS has established a single exception pathway. The CDO (with input from other CMS leadership) has the authority to review and approve exception requests from other federal agencies.

Flexibility in VRDC's Governance and Linkage Architecture

- VRDC accepts requests from a wide array of researchers and research institutions, including graduate students, academics, states, and for-profit entities.
- CMS makes data available for research on many different topics.
- Unlike some other research data repositories, the data in VRDC covers many health conditions and transactions, rendering the data rich and useful to a wide array of researchers.
- CMS provides two options for dataset access (enclave and physical media), which accommodates researchers' differing needs and available resources.
- CMS allows linkage to many different kinds of external datasets, provided the variables needed to link are sufficient and available.

Scalability in VRDC's Governance and Linkage Architecture

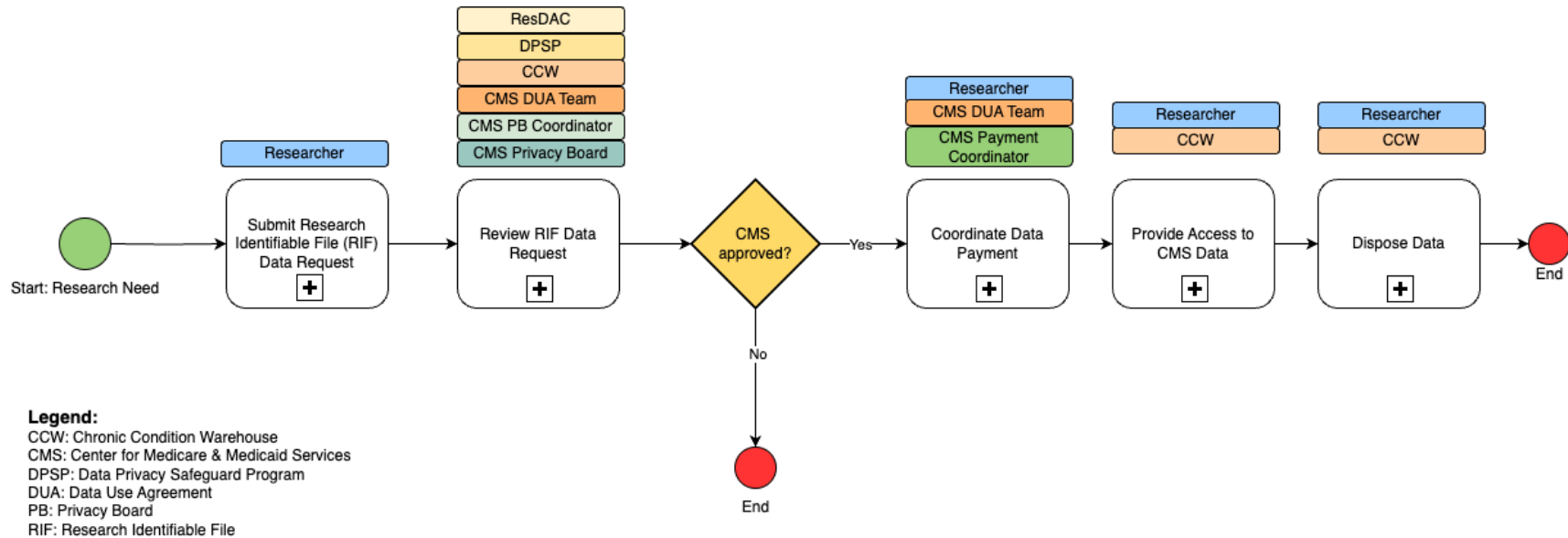
- VRDC's technical infrastructure supports vast amounts of national and longitudinal data, enabling researchers to perform complex, large-scale analyses using advanced computational resources.
- CMS processes hundreds of requests annually, demonstrating its ability to manage high volumes of user activity and data queries effectively.
- There is a decision support tool to ensure researchers are provided with the necessary forms to submit a research request.

- CMS's centralized governance structure, led by the CDO, ensures that scaling decisions are aligned with user needs and strategic priorities, maintaining system stability and operational efficiency.
- CMS recoups its costs for making CMS data available through data fees.

Governance Process

Figure 2 displays the governance process researchers follow to request data from VRDC, largely adapted from OEDA provided business process analysis materials.

Figure 2. Virtual Research Data Center Governance Process to Request Access to a Research Identifiable File



3.3 National Patient-Centered Clinical Research Network

The project team conducted four interviews and reviewed 21 resources for PCORnet (See [Appendix D](#)).

Summary

PCORnet is a network of clinical partners that collaborate to share health data for research.²¹ PCORnet was founded by the Patient-Centered Outcomes Research Institute (PCORI), a nonprofit organization funded through OS-PCORTF, between 2014 and 2017. PCORI carries out and improves comparative clinical effectiveness research (CER) by funding CER research projects and data infrastructure (i.e., PCORnet).²² PCORnet's partnership and operations are informed by three defining characteristics: the network is national in scope, the patient is at the center, and data can be linked to the patient.

PCORnet comprises a coordinating center and eight²³ clinical research networks (CRNs) that share health data as a network of networks (herein referred to as the network).²⁴ Each CRN is composed of multiple health system members that agree to be a part of the CRN and PCORnet. PCORnet represents over 47 million patients across the United States.²⁵ As of 2020, for example, PCORnet CRNs held data from 337 hospitals, 169,695 providers, 3,564 primary care practices, 338 emergency departments, and 1,024 community clinics.²⁶ Some CRNs are formed based on geography (e.g., the Greater Plains Collaborative [GPC] is composed of health systems operating in the Midwest) while other CRNs are based on a shared research interest (e.g., the PEDSnet CRN focused on pediatrics).

PCORnet's technical design is a distributed research network (DRN) where each CRN member health system is a data-contributing site (hereafter "site") that implements the PCORnet common data model (CDM) in a local datamart²⁷ and loads the datamart with patient data extracted from their EHR. A coordinating center creates structured queries, sends them to sites to extract and assemble data for each study, and then compiles data from sites into a single study file for the researcher. Because each site builds and secures its own datamart, patient data are protected behind firewalls and secured based on each institution's preferences. Because data are shared on a study basis, sites maintain control over how their data are used. Over time, site preference has led some CRNs to migrate toward centralized operations where one organization within a CRN holds data from multiple sites.

²¹ PCORnet data sharing is also permitted for select public health, quality improvement, and evaluation activities as well.

²² Between 2010 and 2017, PCORI invested \$325M in starting PCORnet with a majority allocated and spent between 2014 and 2017 when PCORnet officially launched. At this writing, PCORI estimates its current PCORnet support at roughly \$30M per year.

²³ This number has varied over time.

²⁴ PCORnet also formerly supported health plan research networks.

²⁵ PCORnet website: <https://pcornet.org/data/>

²⁶ Forrest, Christopher B., et al. "PCORnet® 2020: current state, accomplishments, and future directions." *Journal of Clinical Epidemiology* 129 (2021): 60-67.

²⁷ A datamart is a subset of a data warehouse or database that focuses on a specific service, department, topical area, or team (<https://www.ibm.com/think/topics/data-mart#>).

PCORnet's governance is tiered so that requirements, responsibilities, and decision making are assigned at the network level, CRN level, or site level. While PCORnet recommends that patients are included at every governance level, it allows each CRN to determine how patient engagement occurs. PCORnet requires sites to build and maintain a datamart, refresh data quarterly, participate in data curation, be responsive in PCORnet processes, participate in studies at their discretion, respond to incoming queries, and abide by PCORnet terms and conditions. Beyond these basic requirements, PCORnet encourages CRN autonomy so that each CRN develops processes and policies that serve its research community. CRNs may instantiate additional policies and processes that help the CRN grow its data and research infrastructure and offerings.

Network governance is overseen by the PCORnet Steering Committee with representation from CRNs and patients. A PCORnet DUA is the primary member policy document that codifies rules for how data is contributed, curated, linked, and used for research. The DUA includes key PCORnet network-wide policies ranging from topics such as prohibitions on reidentifying individuals or sites without written permission, to requirements to acknowledge PCORnet in study-related publications. All sites sign a PCORnet DUA and a CRN DUA. CRN DUAs describe CRN-specific policies, processes, data sharing, and data use. Each CRN has a governance decision-making body of its choosing. CRNs are responsible for patient engagement, such as including patients on committees and in the review process of research projects. For example, the GPC CRN has a Patient Advisory Council that researchers may engage for patient feedback.

Sites are responsible for establishing processes to determine which studies the site will participate in, responding to queries as appropriate, building and refreshing their datamart, and engaging clinical leaders in opportunities for research collaboration. Site governance can include coordinating with the local IRB or other review boards to comply with expectations regarding privacy, security, and human subjects protections. Sites are responsible for managing individual data for patients that wish to be excluded from research or removed from the datamart.²⁸

Research studies using PCORnet data are subject to 45 CFR 46. To align with 45 CFR 46 and ensure the protection of the rights and welfare of human subjects represented in data from PCORnet sites, PCORnet requires that requests for limited or identified data be reviewed and approved by an IRB before a study can begin—usually the local IRB of the lead site or external researcher's organization. Requests for deidentified data do not require IRB review (as they are exempt from the 45 CFR 46) but in some cases, participating sites may request IRB review. Additionally, each site is responsible for taking the appropriate actions with its local IRB for each study in which it participates. By mandating IRB review occur at the site level, PCORnet ensures that each site considers the unique regulatory and ethical requirements of its jurisdiction and community, including the specific risks and benefits to local populations. Additionally, PCORnet maintains active IRB approval for the network infrastructure and operations.²⁹

²⁸ The methods for patients to make such requests are non-standard.

²⁹ Currently provided by Duke University's IRB.

All PCORnet sites are HIPAA covered entities. Therefore, PCORnet and CRN governance complies with the HIPAA Privacy and Security Rules (for example, addressing data retention and destruction of data, and protocols for breaches to align with HIPAA requirements). PCORnet defines limited and deidentified datasets according to HIPAA definitions. Each site's datamart contains dates and selected geographic information and is a limited dataset. PCORnet adheres to the HIPAA Privacy Rule's minimum necessary standard by providing data to researchers that include personal identifiers only when necessary, limiting the research dataset to only the data needed to complete a study, and using aggregate data where feasible.

PCORnet supports research led by investigators at its sites as well as research led by non-PCORnet investigators. That means that a site's PCORnet activities balance supporting research for investigators at their own institution, investigators at other PCORnet sites, and external investigators.³⁰ PCORnet supports observational (data only) and prospective research, including clinical trials, and actively collaborates with industry (e.g., pharmaceutical and medical device companies). PCORnet's prospective research capabilities are one of its most valuable features. Because PCORnet has millions of patients living across the United States, the network is a unique resource to run trials and conduct research on rare conditions or novel treatments where a single institution cannot offer a sufficiently powered research cohort. Under appropriate IRB approvals, PCORnet governance allows sites to share identifiable patient information, contact patients for recruitment, and collect data prospectively.^{31,32}

The coordinating center is an essential role in facilitating PCORnet research. There are two types of coordinating centers: the PCORnet Coordinating Center and a study coordinating center. The PCORnet Coordinating Center processes requests for data, maintains the data model, monitors site data quality, develops and distributes data quality queries, facilitates linkage, and coordinates other network-wide activities. At present, three organizations are contracted by PCORI to support the PCORnet Coordinating Center: Children's Hospital of Philadelphia, Harvard University, and Duke Clinical Research Institute (DCRI).³³ The study coordinating center develops and distributes queries to retrieve data required for a study and then assembles and delivers data to the researcher. Any site can be the study coordinating center and the study coordinating center's costs are baked into the overall cost of the study.

Researchers may request limited datasets, deidentified datasets, or fully identified datasets at the network level (across multiple CRNs) or the CRN level (requesting data only from sites within the CRN). The PCORnet Front Door is the single access point of access for all of PCORnet. Researchers initiate a study by submitting a request to the PCORnet Front Door that the PCORnet Coordinating Center evaluates against basic evaluation criteria (e.g., feasibility, appropriateness, data availability). Researchers generally submit a Front Door request before they have funding to carry out the study and may use a prep-to-research mechanism to gather estimates of the cohort size and gauge sites' interest in participation. This means researchers

³⁰ Requesting researchers are required to acknowledge PCORnet in resulting publications but they choose when and if sites are acknowledged and if sites are included as coauthors.

³¹ Research participants are tagged within the CDM in a Trials table, so that sites maintain awareness about which patients fall under a study-specific research consent.

³² PCORnet does not ingest consent rules into its data model.

³³ All three PCORnet Coordinating Center organizations are affiliated with data-contributing sites.

have not yet completed human subjects protections review by an IRB before they submit their request.

Requests that meet the PCORnet Coordinating Center review criteria are then distributed to CRNs and sites have usually two weeks to opt in or out of study participation (communicated via email). Requests that meet the PCORnet Coordinating Center review criteria are then distributed to CRNs, and sites have usually two weeks to opt in or out of study participation (communicated via email). When appropriate, an information webinar may be held to present the study to interested sites. CRN and site participation is completely voluntary: they are not obligated to share data and many factors could lead to declining. For example, a pediatric CRN would not participate in a study on hypertension in adults >65 years of age. If the PCORnet Coordinating Center determines that a request is best suited for a single CRN, the PCORnet Coordinating Center will deny the request on behalf of the network and redirect the request to the appropriate CRN. Some study requests come directly to CRNs that have their own process for reviewing and approving requests that come directly to their network. Every study requires a study coordinating center to develop queries and compile results. The study's primary investigator (PI) ultimately determines which organization will serve as the study coordinating center.

PCORnet makes data available on a study-by-study basis by securely sending a dataset including the requested data to the requesting researcher. Each study has a dedicated DUA that is created based on PCORnet or CRN DUA templates and specifies what data will be shared, with whom, and for what uses. At times, a contract with DUA terms may be used rather than a DUA to allow terms to be set for funding and data use in one document. Study DUAs limit the researcher's data use to the aims outlined in the DUA, include terms on IRB requirements, mandate registration of the study with ClinicalTrials.gov, and require the researcher to properly acknowledge PCORnet and notify PCORI of all resulting publications. For studies funded by PCORI, researchers additionally complete a Clinical Research Collaboration Agreement that contains a broad statement that requires researchers to adhere to federal, state, and local laws, including the HIPAA Security Rule, informed consent requirements, and regulations of the funding agency.

To link individual data across sites and deduplicate research datasets, PCORnet uses a PPRL solution. PCORnet selected tokenization as its PPRL method and uses Datavant linkage software to generate individual tokens. Site participation in linkage is required. PCORI funds Datavant licenses for the PCORnet Coordinating Centers and sites, but the Datavant license terms limit use of Datavant to PCORnet studies. PCORnet places no additional policies or rules on linked data. PCORnet does not explicitly prohibit linking site data to external datasets. However, external linkage has not yet been attempted. If using PPRL, external linkage would require funds to support a Datavant license for the external dataset.

Linkage is incorporated into the request process. Researchers choose in their request whether or not they want their research dataset to be linked and deduplicated. PCORnet's query process for linkage keeps tokens and linkage maps separate from the clinical data to reduce risk.³⁴ Operationally, this means that one of the PCORnet Coordinating Center organizations receives

³⁴ DCRI handles tokens and matching while Harvard handles clinical data.

and matches tokens and another PCORnet Coordinating Center organization handles the clinical data that queries return from sites. Because PCORnet sites have the same type of data, the aim of linking data is to remove duplicates and connect disparate clinical information for the same patient. However, because few PCORnet sites are geographically close to one another, PCORnet sites might not have many duplicate patients so researchers may not see much difference between unlinked and linked data.

PCORnet data is not free: PIs must compensate sites for participation and the study coordinating center for query development and execution. This cost-recovery model helps sustain PCORnet by recuperating the costs of the network's extensive technical infrastructure and operations. PCORnet has built standard pricing tools but has struggled to find the optimal approach to pricing studies. The study cost can depend on the number of participating institutions, amount of data requested, and addition of linkage, prospective data collection, and site involvement. As part of processing a request, the PCORnet Coordinating Center may provide a range of costs to the researcher after which the PI decides how much funding is ultimately allocated for the sites and study coordinating center.

Case Study from the Greater Plains Collaborative

GPC is one of PCORnet's CRNs. At present, GPC consists of over 10 million covered lives and encompasses over 20 hospitals, 700 clinic locations, and 8,000 providers across the Midwest. GPC is one of several PCORnet CRNs that migrated from a fully distributed model to a more centralized model, with the University of Missouri holding data from multiple GPC sites.

GPC's primary policy document is its DUA for site membership titled *Amended and Restated Greater Plains Collaborative Cooperative Medical Informatics Data Sharing and Network Infrastructure Agreement*. This DUA sets the expectations of participating sites. The GPC DUA is reciprocal, which means that data can move between any sites and any site can serve as a study coordinating center.

GPC has a front porch that is the single-entry point for requests directed to GPC. All requests and subsequent approvals, participation responses, and documentation are stored in a customized RedCap database. A GPC Data Request Oversight Committee (DROC) reviews and approves requests. Each participating GPC site has a primary member and an alternate on the DROC, with patient representation. A majority vote from GPC sites whose data is requested is required for approval. Researchers receiving GPC data for a study sign a GPC Data Transfer and Use Agreement.

Using the PCORnet infrastructure as a foundation, GPC has developed a robust research data ecosystem to drive research at and between its participating sites. GPC is focused on making linkage affordable for all partners and engages in many different types of linkage in addition to the PCORnet PPRL linkage. GPC pursues and secures resources to buy complementary datasets (e.g., death data and consumer data), links them to site data, and makes linked datasets available to researchers for reuse. GPC is pursuing the use of enclaves to provide secure data access through a controlled workspace. GPC has constructed the GPC Reusable

Observable Unified Study Environment (GROUSE) enclave³⁵ that holds a deidentified copy of all participating sites' data and supports two NIH R01 projects.

Limitations in PCORnet Governance & Linkage Architecture

- PCORnet's tiered governance structure can make it difficult to manage consistent adherence to policies and processes delegated to CRNs and sites across a large, federated network. The autonomy granted to each CRN fosters flexibility but can lead to inconsistencies such as differing IRB review processes and data-sharing protocols, especially in multi-institutional studies.
- PCORnet has not implemented external dataset linkage and has used PPRL in a limited capacity. Because many sites are not geographically close, the burden of duplicates and the related value-add from linkage between PCORnet sites is limited.
- Using data from PCORnet is not free. Studies almost always require a grant award or industry support to provide the necessary funding to leverage PCORnet.
- The PCORnet tiered governance means policies and processes can differ between studies, which can be complicated and confusing for researchers and participating sites.
- PCORnet's request and approval process is not fast. Because sites opt in or out of each study, the PCORnet request process includes an extra steps and additional time (i.e., weeks).
- PCORnet's agreement process is not fast. Because PCORnet builds a custom DUA for every study and negotiates the DUA with the researcher's institution, getting to a signed DUA can take months.

Efficiency in PCORnet Governance & Linkage Architecture

- PCORnet is a single infrastructure that supports many different types of research and public health activities by providing health data from over 40 million Americans. This is especially valuable for trials related to rare conditions or novel treatments.
- PCORnet's DRN design keeps the data with or near the organizations that created the data, preserving the knowledge about how that data was created with the organization that holds it.
- PCORnet's tiered governance that assigns decision making to the network, CRN, or site ensures that decisions are assigned to decision makers with the appropriate authority.
- PCORnet's request process allows researchers to gauge interest in study participation and potential cohort site(s), and estimate costs before or as part of writing a grant.

Flexibility in PCORnet Governance & Linkage Architecture

- PCORnet supports research and public health data use across topics and many different types of organizations, including industry partners. Because sites can choose the studies in which they participate, PCORnet does not have to arbitrate requested topics and fit.

³⁵ GROUSE: <https://gpcnetwork.org>

This flexibility also allows the PCORnet technical infrastructure to be well utilized for diverse research purposes by many different organizations.

- PCORnet governance is tiered so that policies and processes are applied at different levels, maximizing CRNs' and sites' autonomy to operate and participate in data sharing in a way that makes the most sense for them.
- PCORnet allows researchers to request data from the entire PCORnet network or a single CRN, so that researchers may conduct a study with the desired size, scope, and geographic reach.
- PCORnet sends data to researchers rather than limiting their access to a controlled environment.

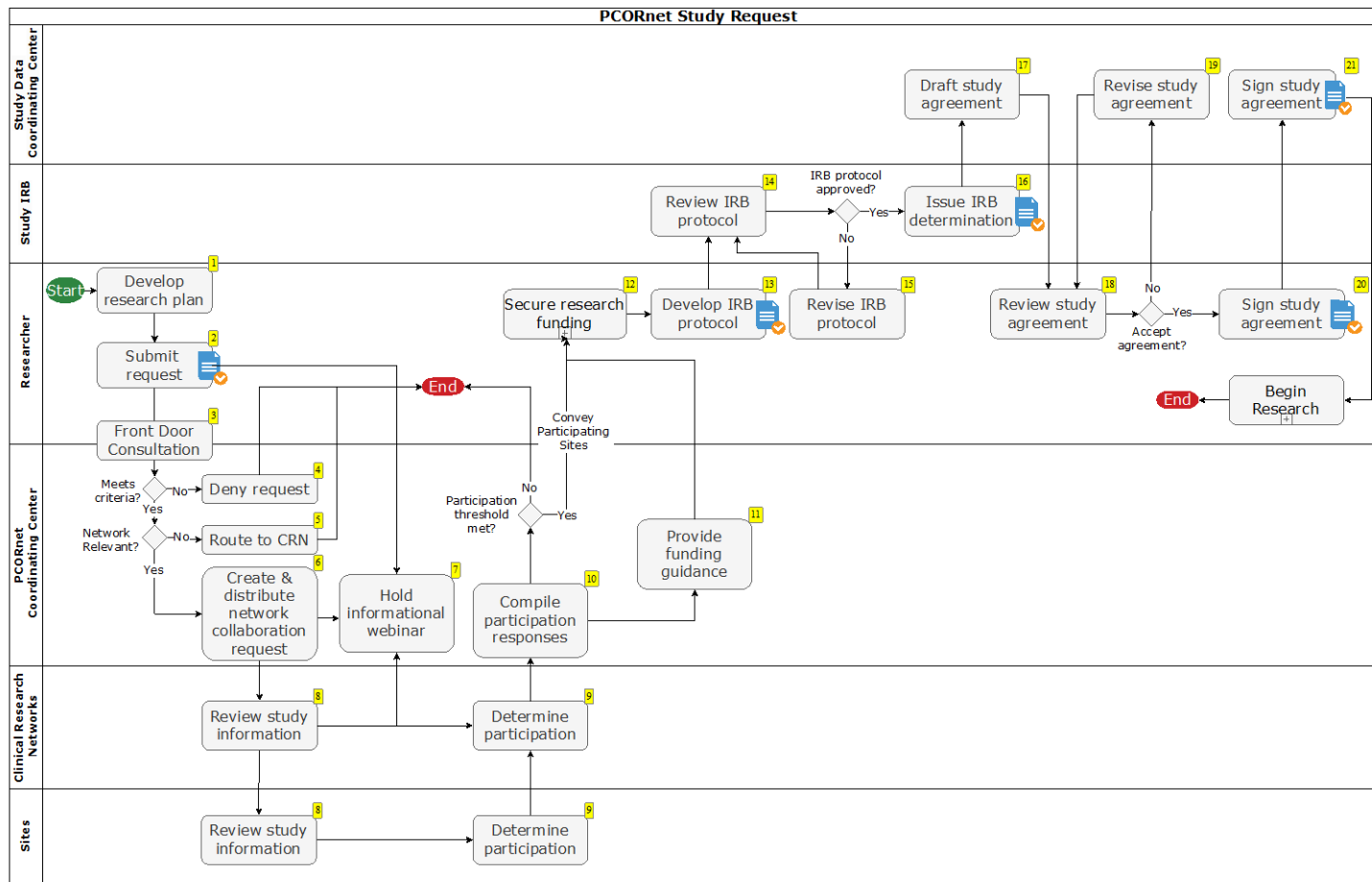
Scalability in PCORnet Governance & Linkage Architecture

- PCORnet governance allows researchers to get study-specific prep-to-research cohort estimates, which helps researchers when applying for grants and funding to support the research study.
- PCORnet releases data on a study basis ensuring sites decide when their data is being used and researchers have access to only the data necessary to conduct their study.

Governance Process

Figure 3 displays the governance process currently used by researchers to request data from PCORnet.

Figure 3. National Patient-Centered Clinical Research Network Governance Process to Request Network Data



4 Targeted Feedback

Targeted feedback discussions with representatives of the broader HHS community highlighted the following points:

- Linkage methods: PPRL is used in the research community but remains experimental; it has not yet evolved to be standardized. As a result, there is not yet consensus on how PPRL linkage should be carried out and whether it is secure and produces an acceptable level of precision and recall. A common data linkage governance framework should allow for linkage methods other than PPRL.
- PPRL linkage technology marketplace: PPRL requires software and the PPRL linkage technology marketplace is rapidly evolving. A few software vendors have emerged as leaders in supporting USG agencies in using PPRL to link datasets. There is concern about the expense of some of these services, and strong incentives and efficiencies to be gained from a unified approach. Because PPRL is relatively new, consensus is needed about the appropriate cost for such services and how to assess PPRL third party software and vendors performance. A common data linkage governance framework should allow for linkage methods other than PPRL and consider how to foster a trusting and fair partnership with software vendors.
- Human subjects protections review: Local IRBs are responsible for evaluating and monitoring the ethics of research studies conducted by researchers within their oversight. Some IRBs have a secondary role as a privacy board and examine whether privacy is sufficiently protected under the proposed linkage approach. This can include visibility into how data from their institution is being used. Thus, local IRBs are the stage-gate to evaluate the ethics and potentially privacy risks with the entire study. A common data linkage governance framework should support coordination with local IRBs according to their norms and practices.
- Token bridging: Token bridging (where tokens are shared across studies and software vendors within a PPRL facilitated linkage) may not always be appropriate. Bridging tokens could jeopardize the intent to ensure that tokens could never reidentify an individual when using PPRL. A common data linkage governance framework should only promote token bridging with a process to evaluate when bridging tokens is appropriate and ethical.
- TEFCA for research: TEFCA provides governance and a technical specification for individual-level linkage allowing health information networks to support health data sharing. TEFCA is standards-based and could not include PPRL in its current state. TEFCA leadership is contemplating a research use case, but TEFCA is not at this time likely to address the challenges that catalyzed this project. A common data linkage governance framework should monitor TEFCA's progress and avoid policies or processes that are not aligned with the TEFCA approach for data linkage and governance.
- Template DUAs: The process for developing agreements can be sped up by starting as a "template" from agreements that have already been reviewed and approved by many attorneys. A common data linkage governance framework should adopt a template

agreement by starting with existing terms and agreement components (including the Common Agreement itself) that have been extensively reviewed and approved.

5 Summary of Findings

The examination of N3C, VRDC, and PCORnet governance models provided insights that merit consideration when creating a common governance framework for linking data.

VRDC's governance model is most applicable to a governance framework for multilateral linkages but does not address all the requirements of linking more than two datasets. As discussed in summaries above, only one of the governance models (VRDC) allows researchers to link data across multiple systems. VRDC's RIF request with external linkage allows researchers to link one dataset of their choosing to CMS data using one CMS-approved linkage method. Many aspects of the VRDC governance model can scale to link three or more datasets, but linking thousands of diverse HHS datasets and data systems introduces unique requirements not relevant to CMS. CMS owns its data and operates a closed system to provision data for research. Because CMS manages requests for a limited number of datasets, their governance does not provide a model for managing scores of potential reviewers across federal agencies. And because CMS restricts linkage to one external dataset and one method, its governance does not make determinations about a proposed linkage implementation or harmonize policies and rules across many systems. While N3C and PCORnet do link many datasets, they also do so using a single method within closed systems so that the governance model does not accommodate governance rules and requirements from many different sources.

Three core roles and functions must be built into a governance framework. All three governance models are grounded in three roles and functions: first, the researcher who is requesting data; second, the coordinator who is navigating the request through the process; and third, the reviewer who approves or denies the request (or requests more information) on behalf of the dataset. In the navigator role, the coordinator serves as the intermediary, communicating with the researcher and reviewer, collecting and storing request-related documentation, and tracking the progress of a request. In the reviewer role, individuals represent their dataset contents, policies, and permitted use. All three functions build toward three key milestones: the submission of the request (which is the unofficial start of the governance process), an approval of the request, and signing an agreement that permits data provisioning and access.

A single front door process streamlines request coordination. N3C and VRDC governance models have a single entry point for data requests. This allows them to have a common and streamlined approach to coordinate the request through a single governance process. PCORNet has a network-wide front door, but it also allows requests to enter at the CRN level. As a result, the PCORnet governance processes and coordination vary depending on where the request is submitted.

A large amount of information is needed to make a decision about linking datasets, and VRDC's submission requirements for external linkage highlight how the amount of information a researcher must provide expands when linkage is proposed. In VRDC's model, the researcher provides a modest amount of additional information to describe the dataset to which they want to link CMS data, the purpose of the linkage, and the variables that

will be used to match individuals. In a scaled environment, where researchers want to link three or more datasets and are proposing a method and novel process to link those datasets, the amount of researcher-provided information would increase even more. Additional datasets and a proposed linkage implementation represent a further expansion of information that a researcher would need to gather, document and provide for decision-making.

Researchers need support to develop a request that is decision ready. All three models acknowledge the importance of helping the researcher build a request that provides the right information for reviewers to make a decision. N3C's request requires the least amount of information, PCORnet's request evolves and expands from the initial request to a more complete study plan, and VRDC's request requires the most extensive and detailed information. Accordingly, VRDC provides the highest level of support. VRDC estimated that its support entity ResDAC can spend upward of 20 hours helping each researcher build a thorough request package with the goal of requiring no additional information to complete their review.

Front-loading study design and IRB work before requesting a data linkage reduces the duration of the governance process. While all three processes start with a request, the work begins before the request when the researcher defines and documents the study design in order to make an informed request. N3C and VRDC governance models require the researcher to complete extensive preparatory and planning activities prior to requesting data. Researchers must design their study, complete IRB review, and submit IRB approvals and supporting documents with their request. While this approach front-loads work for the researcher, it also allows the request to be approved and actioned rapidly. PCORnet allows researchers to submit an initial request without IRB review, enabling them to assess study feasibility, gather demographic data for funding applications, and complete early tasks essential for funding success. Because of this sequence, PCORnet requests experience a pause during the governance process while the researcher completes the IRB process and often applies for grant funding to support their work.

Clarity and direct access to the right reviewer accelerate the speed of review. All three governance models have clear policies about who can approve or deny a request and have built rapid mechanisms to provide the information that a reviewer needs to make a decision. For N3C, two members of the DAC review; in VRDC, review is conducted by members of the Privacy Board that represents all of CMS data; and in PCORnet, all CRNs and sites that intend to participate in the study review study details. All three models conduct reviews asynchronously, making request materials available and collecting responses asynchronously to avoid delays waiting for a review group to convene. N3C also has an escalation process to adjudicate two reviewers that disagree. Reviewers focus on feasibility and permissibility for their data source and rely on the IRB to determine if the proposed research is ethical.

Governance is most efficient when committing to only one method of linkage. N3C, VRDC, and PCORnet governance each permit only one linkage method and allow only one entity to perform that linkage. For example, VRDC allows only clear text linkage based on a limited set of VRDC-specified identifiers and allows just one CMS-contracted organization to perform the linkage. Limiting linkage to one method and linkage partner means that reviewers do not need to contemplate novel or unfamiliar linkage methods, vet various organizations that

will handle data during linkage, or estimate whether the proposed linkage approach is secure and likely to be successful.

The models do not apply additional restrictions and policies to linked data. The three models only follow the rules of the originating datasets, without adding extra rules or restrictions, making linked data easy for researchers to use and access. The governing rules remain focused on data sensitivity and the amount of PHI present in the dataset to which the researcher will gain access. In VRDC, individual linked data is categorized in the higher level of sensitivity (i.e., RIF), recognizing that linkage requires sharing identifiers and could increase risk of reidentification.

Non-negotiable DUAs streamline and accelerate execution of data sharing agreements. N3C and VRDC use a non-negotiable DUA approach with a modestly customized standard DUA template; the DUA terms may not be contested or modified by the researcher's institution. In VRDC's case, the researcher signs the DUA before the request is even approved. Alternatively, PCORnet builds a new agreement for each study that is negotiated with multiple attorneys that can take months to execute. An added benefit of a non-negotiable DUA is that the rules for using datasets from N3C and VRDC never vary as the terms of use across all DUAs are the same.

An exception pathway adds flexibility to a governance framework. Innovative study methods and topics can push the boundaries of existing data use policies, and in these instances, researchers would need an exception for the study to be permissible. N3C's governance model does not allow researchers to request exceptions to any rules or terms but VRDC's governance model provides an exception process for some types of research. Exceptions did not surface in the PCORnet governance model likely because this model is fully customizable and has limited standard policies that apply writ large. When and how exceptions are requested and processed is an important consideration for the common governance framework.

Governance process steps were similar, but their sequence varied across models. All three models are defined by similar steps: study design, IRB review and approval, request, review and approval, data use agreement, and payment (if required). In some instances, study design can include securing grant funding. N3C and VRDC expect study design, IRB, and agreement to happen before the request so that moving from approval to study start and data access can happen quickly. Unlike VRDC and PCORnet, which involve payment processes, N3C offers free data access, eliminating payment steps and saving time. Researchers seeking VRDC and PCORnet data often need grant funding, which increases the overall time investment.

6 Conclusion

The project team assessed three governance models to identify effective policies, processes, and practices for creating a unified framework that fosters multilateral connections. Governance models from N3C, VRDC, and PCORnet were evaluated for efficiency, flexibility, and scalability to create multilateral linkages for research. All three models have the same general process steps, roles, and key milestones (e.g., submitting a request, receiving an approval from the IRB,

conducting a review, and executing an agreement) but carry out those steps in different ways. Each model takes a different approach to linking data, with only VRDC providing an applicable governance model for linking across multiple systems. All three models highlighted best practices to inject efficiency and flexibility into a governance framework and offered approaches, tools, and learnings to inform a common governance process.

Appendix A. Data Collection Tool Questions

The project team developed the following questions for the data collection tool, to be used during the research process and for interviews.

Overview

- **Technical Overview**
 - What is the technical design?
 - What is the high-level technical architecture of the data-sharing arrangement?
 - How many organizations contribute data to the sharing agreement?
 - What types of organizations contribute data?
 - What types of data are shared?
 - What amount of data is available?
 - What are the modes of data access?
- **Governance Overview**
 - What is the overall governance approach and management infrastructure?
 - What are the policies and rules regarding data contribution?
 - Is there a participant or member agreement?
 - Who are the decision-making bodies?
 - What are the policy documents and what topics do they cover?
- **Data Use**
 - Does the model permit secondary data use?
 - Are there policies on secondary data use?
 - Are there limitations on secondary data use?
 - Are there types of secondary data users?
 - What are the number of studies/requests to use data?
 - What types of research is this system supporting?
 - How does the model consider data use preferences from patient consents for research?
 - Can the model remove patients who withdraw consent?
 - How does the model monitor and enforce data use restrictions?
- **Linkage and Linked Data**
 - Is the data in this model already linked (hashes or longitudinal records)?
 - What are the linkage and technology methods used?
 - Can researchers ask to link data from this model to external datasets?
 - How can researchers link data from this model to external datasets?

- What methods of linkage are permitted (e.g., PPRL, clear text)?
- What policies and rules apply to the data linkage process and linked data?
- How are researchers able to access linked data?
- What policies address linked data?
- What limitations does the organization place on the purpose of linkage?

Governance for Requesting Data for Linkage

- ***Linkage Request Review and Approval***

- How do researchers submit a request for a data set for linkage (e.g., portal, forms)?
- What information does the request require?
- Who or what entities review the request?
- What are the criteria for assessing the request?
- Does the researcher need to establish access with any platforms to submit the request or receive data?
- Does the researcher need to review any documents (e.g., master policy document, researcher code of conduct)?
- Do data contributing sites need to opt in to participate in a linkage study? If so, describe the process.
- Does a researcher need to take any training before the linkage study can begin?

- ***Human Subjects Protections***

- What are the human subjects review requirements?
- Are there special human subjects requirements for linking data sets?
- How do researchers deal with multiple data set owners?
- How many resubmissions to IRBs generally occur?
- What are the human subjects-relevant rules or policies that the ethics review imposes on researchers requesting data?
- How are consent processes handled for data subjects in longitudinal studies or changes in data use over time?

- ***Privacy Board, Privacy and Security***

- What are the privacy board review requirements for data contribution?
- What are the privacy board review requirements for data request linkage?
- What must the researcher submit for privacy board review?
- What are the standards the researcher must meet for privacy?
- How many resubmissions to the privacy board(s) generally occur?
- What are the privacy and security rules or policies imposed on researchers requesting data for linkage (e.g., enclaves and data release processes)?

- What monitoring and auditing mechanisms are in place to ensure ongoing adherence to privacy protocols?
- What training is provided to researchers to ensure understanding of and adherence to privacy, security, and ethical standards?
- What are the policies to ensure privacy and security through the linkage process?
- What are the approaches and governance to ensure security?
- **Data Use Agreements**
 - Does a researcher requesting data need to complete a Data Use Agreement?
 - What is the name of this agreement?
 - Is there a template?
 - What parties sign this agreement?
 - What parts of the DUA are negotiable?
 - What are the rules or policies that the DUA imposes on researchers requesting data for linkage?
 - Do the terms of the DUA address:
 - Initial location of the data?
 - Permissible technologies used to transfer the data?
 - Permissible storage locations for the data?
 - Whether the data are to be accessed only within an enclave owned and operated by the data owner?
 - Rights in data/research results (including any requirements of acknowledgement of data providers)?
 - Data subject consent?
 - Data fields that may be/are included within data sets?
 - Prevention of re-identification and specific restrictions on contacting data subjects?
 - Privacy and security training and education requirements?
 - Scope of use:
 - Governed by a submitted protocol?
 - IRS/privacy board/approval required?
 - Project is specific/scoped?
 - Linkage permissible/linkable data sets specified?
 - Permissible data sharing/disclosure?
 - Data retention/destruction requirements?
 - Requirements related to breach:

- Notification to data originator?
- Notification to data subjects?
- Notification to law enforcement?
- Financial responsibility in the event of a breach?
- How many DUA iterations generally occur?
- Are there other agreements the researcher must complete?
- Are there other notable clauses of the DUA that could be relevant to other government-sponsored large data sharing initiatives for health research?

Applicable Laws and Policies

- What laws apply to the data in this model?
- What are the requirements from these laws with which the model complies?
- How does the model comply with those requirements? For privacy and security standards/controls, does the repository use or adapt a standard framework? Is it tailored? Does it seek exceptions to security standards? What is the process for seeking exceptions?
- What is the model's deidentification standard?
- How does the organization ensure compliance with new data protection laws and regulations or applying new laws to emerging technologies?

Other Governance

- Are there any exceptions?
- Are there any forms of payment?

Qualitative Questions

- Is the governance model supporting data use working as well as expected?
- What key values informed your governance approach?
- Where do linkage requests get "stuck"?
- What are the most difficult barriers to linking datasets outside (e.g., challenges associated with IRB review, addressing privacy requirements, or executing the DUA)?
- What are solutions the model created to overcome these barriers?
- What are broader changes that would enable streamlined data linkage?
- What are near- and mid-term plans to modify governance?
- What are long-term plans to modify governance?
- What does a governance framework need in order to move quickly and to be flexible and scalable?
- How is the model funded? What are the greatest expenses?
- What were/are the critical partnerships establishing and managing governance?
- How have patients been included in the governance?
- What are/were the critical partnerships establishing and managing governance?

Appendix B. National Clinical Cohort Collaborative Materials Reviewed

The project team reviewed 35 resources for N3C. Citations for the three interviews and materials are shown below.

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Appendix C. Virtual Research Data Center Materials Reviewed

The project team reviewed 29 resources for VRDC. Citations for the two interview and materials are below.

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The project team reviewed 21 resources for PCORnet. Citations for the five interviews and materials are below.

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